

Controllers for a prototype/MVP level application

If your product is in the R&D phase, wherein your team is working on prototypes to generate a minimum viable product (MVP), you can consider some development boards that are suitable for prototyping. Later, once you have established the product's features, you can move on to a more specialised controller.

In fact, vendors like Adafruit actually offer ready-to-use microcontroller platforms that you can use to build a wearable device. The Adafruit FLORA platform is one such example. It's a circular, sewable microcontroller that works with Arduino. The LilyPad Arduino is a similar open source platform. Using conductive thread, it can be attached to fabrics and linked to other electronic components, and hence, it is suitable for e-textiles.

ing functionality are slowly becoming common, and a lot of research is going on in this arena. For example, the Advanced Technology Institute (ATI) in Surrey has shown how their innovative photo-rechargeable technology, which combines perovskite solar cells and zinc-ion batteries, could enable wearables to be powered with just a few seconds of sunshine.

The spread of wearable integration. Wearables can be a discrete product, but at the same time they can be integrated into existing serv-

es. The most famous example of this is probably Google TV. Google plans to incorporate fitness trackers into Google TV and Android TV. The purpose is to enable users of Fitbit and Wear OS to display real-time metrics like heart rate and calories burned.

E-tattoos, stickers, and implantables. E-tattoos hold enormous potential, particularly for the medical field's use in blood pressure monitoring. Researchers from the University of Texas at Austin have published a paper outlining e-tattoos and how

they can monitor blood pressure, something that conventional wearables normally cannot. The team created a custom PCB with the ARM Cortex M4 microcontroller to interact with the tattoo. Their findings demonstrate the vast potential of e-tattoos.

Smart clothing. Just like e-tattoos, e-textiles are also rapidly becoming popular. Many researchers are working on it and, over time, several use cases have sprung up. Even the National Basketball Association in North America is looking into deploying wearable e-fabrics to prevent player injuries. Nextiles, a materials science company, has developed socks that could be used to gather and analyse data about injuries to the ankle. **EFY**

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